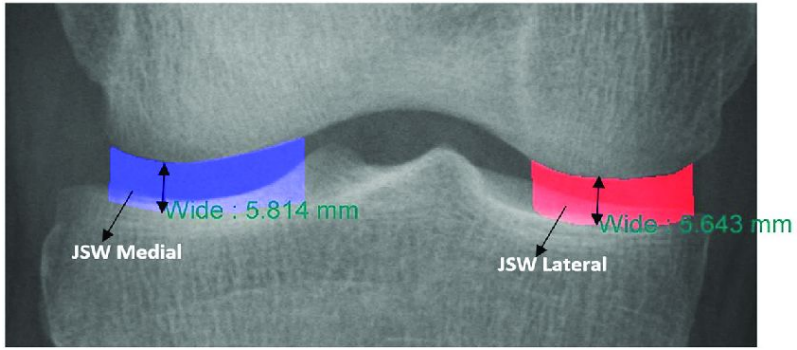
Introduction:

$$Y = \sum_{j=1}^{m} X_j \theta_j + \gamma Z + \varepsilon$$

X: AUC cutting
Z: Not defined
Y: Medial Minimum Joint Space Width



Y selection:

Contents of kxr_qjsw_duryea08

V08MCMJSW	Num	8	BEST12.	BEST32.	BL/FU kXR reading (JD): medial minimum JSW [mm]

---**Joint space width (JSW)**: measured in millimeters (mm), represents the minimum distance between the femur and tibia within the knee joint.

~A **larger** JSW indicates healthier cartilage and less structural degeneration.
Hence, if $\beta > 0$: An increase in activity is associated with an increase in JSW, suggesting a protective effect on joint structure.

~A **smaller** JSW reflects cartilage loss and narrowing of the joint space.
Hence, if $\beta < 0$: An increase in activity is associated with a decrease in JSW, suggesting a potentially harmful effect or accelerated joint space narrowing.

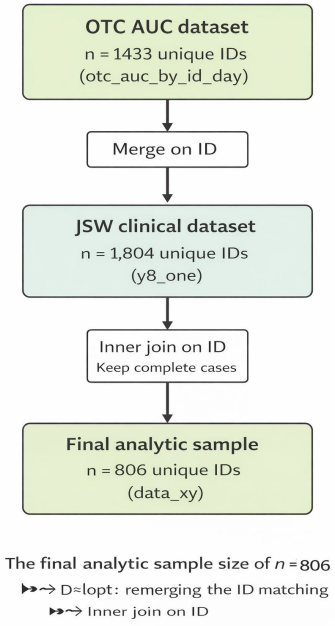
Data:

```
#read data
wd <- "/project/rrg-oespinga/oespinga/OAI/ClinicalData/CompleteData_ASCII/"
min8=read.table(paste0(wd,"/Acceldatabymin08.txt"), sep="|", header=T)
```

```
####PUT IN Y
y8=read.table(paste0(wd,"/kxr_qjsw_duryea08.txt"), sep="|", header= T)
```

	ID	Y	Z	AUC_1	AUC_2	AUC_3	AUC_4	AUC_5	AUC_6	AUC_7	...	AUC_52	AUC_53	AUC_54	AUC_55	AUC_56	AUC_57	AUC_58	AUC_59	AUC_60	n_minutes
0	9000099	3.800	0	59.375000	37.847222	28.521825	24.007937	21.527778	20.684524	20.585317	...	0	0	0	0	0	0	0	0	0	10080
1	9001695	3.891	0	73.561508	39.434524	23.809524	14.484127	10.119048	7.589286	5.257937	...	0	0	0	0	0	0	0	0	0	10080
2	9001897	3.048	0	11.855159	1.488095	0.347222	0.000000	0.000000	0.000000	0.000000	...	0	0	0	0	0	0	0	0	0	10080
3	9002116	3.785	0	34.953704	14.930556	6.211420	3.163580	1.967593	0.925926	0.617284	...	0	0	0	0	0	0	0	0	0	12960
4	9002817	1.628	0	40.029762	17.807540	10.267857	7.936508	4.811508	2.281746	0.744048	...	0	0	0	0	0	0	0	0	0	10080
5	9003406	3.531	0	3.521825	1.835317	0.992063	0.694444	0.347222	0.297619	0.148810	...	0	0	0	0	0	0	0	0	0	10080
6	9006407	4.597	0	21.279762	10.615079	6.001984	3.918651	2.331349	0.892857	0.148810	...	0	0	0	0	0	0	0	0	0	10080
7	9006723	4.031	0	46.279762	18.700397	8.184524	5.605159	4.265873	1.785714	0.000000	...	0	0	0	0	0	0	0	0	0	10080
8	9008934	5.600	0	60.069444	26.388889	9.523810	2.976190	1.289683	0.446429	0.248016	...	0	0	0	0	0	0	0	0	0	10080
9	9009927	4.011	0	47.371032	20.734127	8.928571	5.357143	3.769841	3.025794	1.736111	...	0	0	0	0	0	0	0	0	0	10080

	ID	Y	Z	AUC_1	AUC_2	AUC_3	AUC_4	AUC_5	AUC_6	AUC_7	...	AUC_52	AUC_53	AUC_54	AUC_55	AUC_56	AUC_57	AUC_58	AUC_59	AUC_60	n_minu
count	8.040000e+02	804.000000	804.0	804.000000	804.000000	804.000000	804.000000	804.000000	804.000000	804.000000	...	804.0	804.0	804.0	804.0	804.0	804.0	804.0	804.0	804.0	804.0000
mean	9.493877e+06	3.919338	0.0	42.714310	19.474127	10.009207	5.774458	3.611806	2.347848	1.580207	...	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	10397.7101
std	2.911380e+05	1.266116	0.0	21.829213	13.979507	9.223170	6.438571	4.844019	3.875737	3.249930	...	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	1904.3433
min	9.000099e+06	0.000000	0.0	0.000000	0.000000	0.000000	0.000000	0.000000	0.000000	0.000000	...	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	1440.0000
25%	9.240515e+06	3.154250	0.0	26.911630	9.255642	3.323413	1.140873	0.347222	0.099206	0.000000	...	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	10080.0000
50%	9.486578e+06	4.000500	0.0	39.405909	16.035604	7.249504	3.565228	1.540454	0.595238	0.151565	...	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	10080.0000
75%	9.742812e+06	4.705500	0.0	55.567956	26.500496	14.310516	8.246528	4.861111	2.790179	1.587302	...	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	10080.0000
max	9.999878e+06	8.000000	0.0	145.152324	120.635980	90.282410	57.204803	39.880952	39.434524	39.236111	...	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	36000.0000



The final analytic sample (n = 806) was obtained by merging the accelerometry AUC dataset (n = 1433 unique participants) with the clinical JSW dataset (n = 1804 unique participants) using participant ID. Only individuals present in both datasets were retained.

Variable:

How physical activity distribution is associated with structural knee health?

ID:Unique participant identifier

Y (Outcome Variable): Medial minimum joint space width (JSW), measured in millimeters (mm)

AUC_1 – AUC_{60} : These are the key exposure variables. Each AUC_j represents: The contribution to the Occupation Time Curve (OTC) at activity intensity threshold j.

Z (Group Indicator)

Model:

$$Y = \sum_{j=1}^{m} X_j \theta_j + \gamma Z + \varepsilon$$

```
Restricted license - for non-production use only - expires 2027-11-29
Gurobi Optimizer version 13.0.1 build v13.0.1rc0 (win64 - Windows 11.0 (26100.2))

CPU model: Intel(R) Core(TM) i9-14900HX, instruction set [SSE2|AVX|AVX2]
Thread count: 24 physical cores, 32 logical processors, using up to 32 threads


---------------------------------------------------------------------------
GurobiError                               Traceback (most recent call last)
Cell In[5], line 57
     54     model.addSOS(GRB.SOS_TYPE1, [y2[t], k2[t]])
     55     model.addSOS(GRB.SOS_TYPE1, [y3[t], k3[t]])
---> 57 model.optimize()
     59 beta1 = beta[0].x
     60 beta2 = beta[1].x

File src/gurobipy/_model.pyx:926, in gurobipy._model.Model.optimize()

GurobiError: Model too large for size-limited license; visit https://gurobi.com/unrestricted for more information
```

The error message "**Model too large for size-limited Gurobi license**" or "**Model too large for restricted Gurobi license**" is displayed when Gurobi uses a restricted, size-limited license, and the model size exceeds these limits, *i.e.*, 2000 variables, 2000 linear constraints, and 200 variables if quadratic terms are present.

```
df = pd.read_csv("/project/def-oespinga/xiankai/oai_rep2_m50.csv")

df = df.iloc[:100, :]

data = df.to_numpy()

Y = df["Y"].to_numpy().reshape(-1, 1)
Z = df["Z"].to_numpy().reshape(-1, 1)

auc_cols = [c for c in df.columns if c.startswith("AUC_")]
X = df[auc_cols].to_numpy()

n = X.shape[0]
m = X.shape[1]
auc_cols = [f"AUC_{i}" for i in range(1, 22)]
```

cut row by first 100

cut row by first 22

Result:

```
beta1: 0.1592326751079391
beta2: -0.18457757709814976
beta3: 0.014159414267011942
jump1: 1.0
jump2: 2.0
gamma: 0.0
```

Breakpoints (Jump1 = 1, Jump2 = 2):
The model identifies structural changes in the very low-intensity activity region, suggesting that associations with JSW are concentrated at the lower end of the activity distribution.

Segment 1 (β_1 = 0.159):
In the lowest intensity range, greater activity mass is positively associated with medial JSW, indicating a weak protective relationship.

Segment 2 (β_2 = −0.185):
Immediately after the first breakpoint, the association becomes negative, suggesting a potential adverse effect in this narrow intensity band.

Segment 3 (β_3 = 0.014):
Beyond this region, the slope is near zero, indicating minimal association between higher intensity activity and JSW.

Next Step:

1. Getting the Gurobi license to conduct the full model.

2. Connecting the cutpoint, NA values or other setting with team member.

3. Trying other possible Y variable and maybe finding appropriate Z variable.

4. Fully understanding the model (better for the report).

5. To be continued…